

## Comparing ODE Models of Chimeric Antigen Receptor T cell therapy

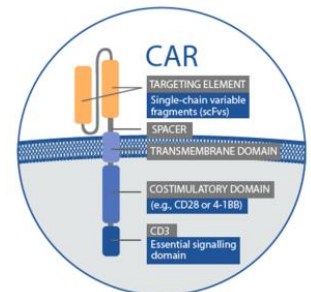
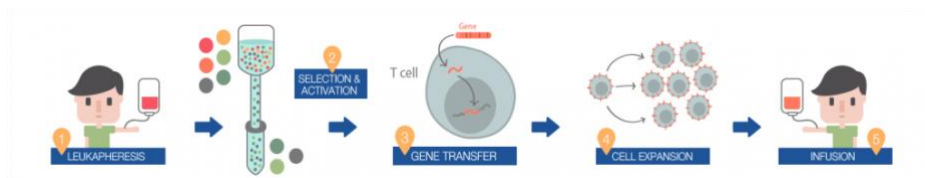
### **Project Steps:**

1. Read/Discuss Owens & Bozic BMB paper
2. Write python code to reproduce figures from Owens and Bozic
3. Read Liu et al paper with the goal of understanding where the data comes from, what are the units and how can we relate them to output from our model?
4. Write python code for loading the data, plotting and performing some statistical analysis.
5. Parameter estimation via least squares in python
6. Intro to mixed-effects population modeling in monolix
7. Download monolix and set-up
  - a. Data file from Liu et al
  - b. Model file from Owens and Bozic
8. Set up project and run to estimate parameters for the Owens and Bozic model\*

\*Maybe try a simpler model with fewer parameters first (Sahoo et al.)

### **Biological Background:**

- Chimeric antigen receptors (CAR) are engineered fusion proteins that target T cells to a specific antigen present on tumor cells to generate an antitumor immune response.
- The CAR protein contains an extracellular domain which binds to a surface marker expressed on tumor cells (and ideally not healthy cells), making them specific. It also includes an internal co-stimulatory domain which allows the engineered T cells to mount an immune response after binding to their target antigen, bypassing the requirement of additional, external costimulation inherent to endogenous T cells.
- Brief summary: A patient's T cells are isolated from a blood sample. A viral vector is used to insert the CAR gene into their T cells, which are then expanded ex vivo. Patients receive a round of lymphodepleting chemotherapy to reduce their endogenous T cells population and thereby make room for the CAR T cells to expand. Finally, CAR T cells are infused into the patient.



- The earliest CAR T cell technology was actually developed to try to cure HIV; however, this has proven a challenging area of application due to the low levels of antigen expression in controlled HIV infections (among other reasons).

- The first success came in treating blood cancers. CAR T cells targeting CD19, which is expressed on malignant B cells, were found to have potent activity in early phase clinical trials involving patients with relapsed and/or refractory B cell malignancies over a decade ago now.
- Long-term follow-up data indicate that CD19-targeted CAR T cells are likely to be curative for a subset of patients with B cell lymphomas.
- CD19-targeted CAR T cells are now approved for relapsed and/or refractory B cell lymphoma and B cell acute lymphoblastic leukaemia, and B cell maturation antigen-targeted CAR T cells are approved for relapsed and/or refractory multiple myeloma.
- CAR T cell therapies have become an important component of the treatment landscape for a range of haematological malignancies, and the indications for these therapies continue to expand to earlier lines of treatment.

| Disease                           | Target | Response rate | FDA-approved drugs                                                                           |
|-----------------------------------|--------|---------------|----------------------------------------------------------------------------------------------|
| Large B-cell Lymphoma             | CD19   | 40-74%        | axicabtagene ciloleucel (2017)<br>Tisagenlecleucel (2018)<br>lisocabtagene maraleucel (2021) |
| B-cell Acute Lymphocytic Leukemia | CD19   | 71-81%        | Tisagenlecleucel (2017)<br>brexucabtagene autoleucel (2021)                                  |
| Multiple myeloma                  | BCMA   | 73-98%        | idecabtagene vicleucel (2021)<br>ciltacabtagene autoleucel (2022)                            |

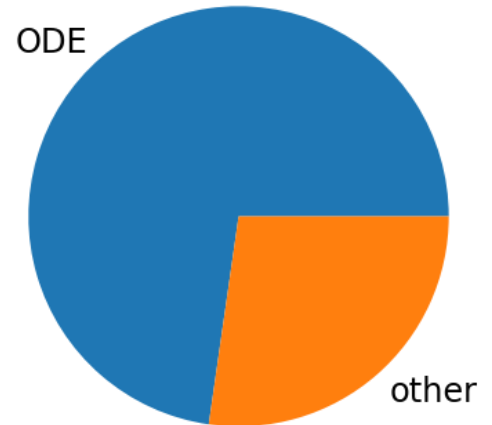
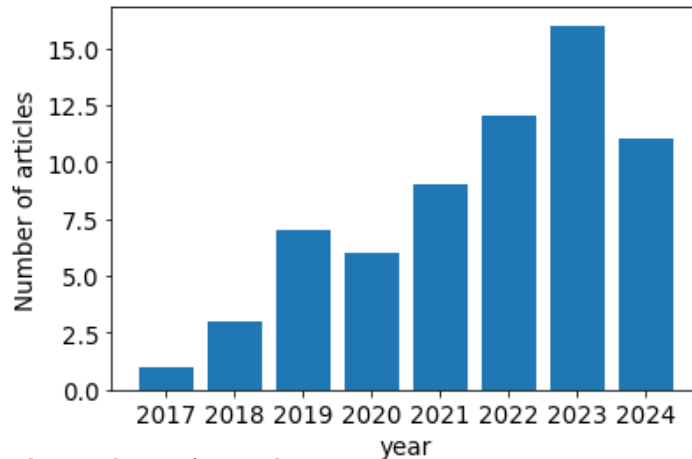
Cappell and Kochenderfer 2023

- There is also a lot of interest in extending this technology to treat other malignancies and other diseases, including HIV.

#### **Mathematical Modeling:**

- Mathematical modeling of CAR T cell therapy is a growing area of work. I queried google scholar for articles with “CAR T cell” and either “mathematical” OR “quantitative” OR “math” OR “model” in the title to generate the bar chart below.
- Most models are ODE-based. A subset of these have been fit to data.
- Existing studies address diverse scientific goals:
  - Optimize design or manufacturing process
  - Elucidate CAR T-cell proliferation and/or killing mechanisms
  - Predict patient outcomes
  - Suggest optimal treatment plans

- Increasingly there is data tracking CAR T cell counts and tumor size, even CAR T cell in vivo imaging tracking location of CAR T cells so I think this is an exciting area of application for data driven modeling.



#### Open Questions/Challenges:

- CAR T cell treatment can fail if **tumor heterogeneity** leads to antigen escape. We could build a model that includes a resistant subpopulation and look into evolutionarily informed treatment plans with pulsed CAR T cell therapy. (To manage exhaustion/toxicity people have developed CAR T cells with on/off switches).
- For solid tumors, achieving **adequate T cell trafficking to the tumor site and tumor infiltration** is challenging. We have a simple spatial model for local delivery which we could use to model additional treatment plans including multiple doses, tumor resection + local delivery, time-release delivery.
- Managing toxicity, minimizing risk of post-treatment infections.** Consider modeling combinations of pre-conditioning + CARs + drugs for managing CRS.
- Immunosuppressive tumor microenvironment.** Include more careful model of tumor-immune interactions with the endogenous immune system and CAR T cells including phenomena like exhaustion, bystander killing.
- What is the optimal balance of CD4 vs CD8 CAR T cells, memory vs naïve etc.

#### Motivation:

Before diving into these questions, it would be great to go back to the collection of ODE models that have been developed for CAR T cell therapy, and calibrate a subset of them to data on tumor size and CAR T cell count from B cell malignancies. This could serve as a literature review + model comparison. It would allow us to pick a good mechanistic model for future work by identifying which model (or collection of models) best explain available data.

#### Relevant CAR T cell Model Reading:

- Simple model fit to in vitro data: Sahoo et al. 2020  
[https://royalsocietypublishing.org/doi/full/10.1098/rsif.2019.0734?rfr\\_dat=cr\\_pub++0pubmed&url\\_ver=Z39.88-2003&rfr\\_id=ori%3Arid%3Acrossref.org](https://royalsocietypublishing.org/doi/full/10.1098/rsif.2019.0734?rfr_dat=cr_pub++0pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org)
- Our model not fit to data, which I would like to validate against data: Owens and Bozic 2021 <https://link.springer.com/article/10.1007/s11538-021-00869-5>
- Liu et al. 2020 digitized data on CAR T cell kinetics from 217 patients across 7 trials and compared kinetic parameters when stratified by response status, patient populations, and tumor types. They used monolix to fit their piecewise model—we could take a similar approach with mechanistic models.  
<https://ascpt.onlinelibrary.wiley.com/doi/full/10.1002/cpt.2040>.
- Liu et al. 2022 (different Liu than above!) I haven't read this in detail yet, but they provided the most ready-to use data set which has both tumor and CAR T cell data over time. "Real-time CAR T-cell and tumor burden data of 209 patients were collected from clinical studies and standardized with unified units in bone marrow."  
[Liu et al. 2022 Computational model of CAR T-cell immunotherapy dissects and predicts leukemia patient responses at remission, resistance, and relapse.](#)
- More recent model accounting for different CAR T cell phenotypes fit to clinical data: [Kirouac et al. 2023](#), and [a follow up from the same author in 2024](#)

#### Available Data:

- CAR T cell kinetics + patient response: Liu et al. 2020 developed a cellular kinetic model to retrospectively **characterize CAR-T kinetics** in 217 patients from 7 trials and compared CAR-T kinetics across response status, patient populations, and tumor types. Model consists of linear kinetics applied in phases. They track effector and memory subsets of T cells. The data they used is available as supplemental info here:  
<https://ascpt.onlinelibrary.wiley.com/doi/full/10.1002/cpt.2040>.
  - o One paper they cite is the OG Tisagenlecleucel kinetics:  
<https://ascpt.onlinelibrary.wiley.com/doi/10.1002/psp4.12388>
  - o Last page of the supplement has Tisagenlecleucel + tumor data:  
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7013261/>
- CAR T cell kinetics and tumor data: [Liu et al. 2022 Computational model of CAR T-cell immunotherapy dissects and predicts leukemia patient responses at remission, resistance, and relapse.](#) Data is on their github page!  
<https://github.com/Lunan12/CAR-T-model>
- Other models fit to data from both CAR T cells and tumor, which we could request/digitize:
  - o Kimmel et al. zuma data:  
[https://royalsocietypublishing.org/doi/full/10.1098/rspb.2021.0229?rfr\\_dat=cr\\_pub++0pubmed&url\\_ver=Z39.88-2003&rfr\\_id=ori%3Arid%3Acrossref.org](https://royalsocietypublishing.org/doi/full/10.1098/rspb.2021.0229?rfr_dat=cr_pub++0pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org)
  - o Model fit to data including endogenous T cells:  
<https://aacrjournals.org/cancerrescommun/article/2/11/1532/711340/Mechanistic-Modeling-of-the-Interplay-Between-Host>

- [Effector, Memory, Exhausted cell model](https://www.nature.com/articles/s41591-018-0010-1). (Kirouac et al) We could digitize data ourselves from the same source, which is Figure 1 here:
- Another recent model from PKPD perspective to consider: <https://www.nature.com/articles/s41540-024-00355-3?fromPaywallRec=false>
- Ahmed et al. fit to in vitro data from cell assays + clinical data <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10508581/>
- Look at models fit to data in Nukula et al. review—particularly if individual-level data has CAR and tumor:
  - Nukala et al: <https://arxiv.org/pdf/2101.05359>
  - [Tserunyan et al. is a little less helpful, but includes some more recent papers.](#)
- In vitro experiment elucidating killing mechanism: <https://www.nature.com/articles/s41419-024-06461-8>
  - And a relevant modeling paper: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9983577/>

### Data Fitting Options:

- Individual-based approach
  - Minimize RSS between data points and model prediction for each individual
  - Does not incorporate information about parameter values across individuals
  - Assumes gaussian distributed noise on each data point.
  - Use built-in function in python, matlab, R or other
- Population-based approach
  - Population mixed effect model. Measurement from individual  $i$  at timepoint  $k$  modeled as  $y = f(t, \theta) + \epsilon$ , where  $f$  represents the solution of the ODE model for the relevant state variable,  $\theta$  is the parameter vector for individual  $i$ , and  $\epsilon \sim N(0, \sigma^2)$  is the measurement error for the data.
  - We have a statistical model that includes the mechanistic model, an error model, and any possible correlations/covariates in the population
  - Use monolix